Metallic Materials for Biomedical Applications

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Choice of Implant material

Requirement for the choice of the biomaterial:

- Acceptability by the human body

- Should possess sufficient mechanical strength to sustain the forces to which they are subjected

- Should have very high corrosion and wear resistance

A biomaterial should remain intact for a longer period (15 to 20 years)

Choice of bioimplant/biomaterial…..

The success of a biomaterial or an implant dependents on

- The properties (mechanical, chemical and tribological) of the biomaterial in question
- Biocompatibility of the implant
- The Patient-specific requirements

Biocompatibility

- A biocompatible material disrupts the normal body function as little as possible
- Factors which influence implant biocompatibility; implant size, shape, material
- composition, surface wettability, surface roughness and charge.

Biocompatibility….

The biomaterial must not:

- Change plasma proteins (including enzymes) so as to activate adverse reactions
- Cause thrombus formation, adverse immune response or cancer
- Destroy or sensitize the cellular elements of blood
- Produce toxic or allergic responses
- Deplete electrolytes
- Be affected by sterilization

Biocompatibility…..

- The environment should not cause
  - Degradation (biological or mechanical), if required
  - Corrosion of the biomaterial

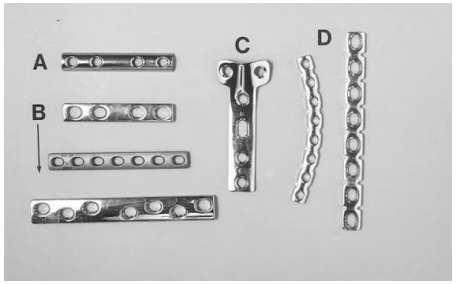
- In practice, no synthetic material is completely harmonious with the living environment

Bioimplants

- Materials used to fabricate bioimplants ;
  - Metals
  - Alloys
  - Polymers
  - Ceramics

- These implants encounter different biological environments of very different physico-chemical nature

Bioimplants

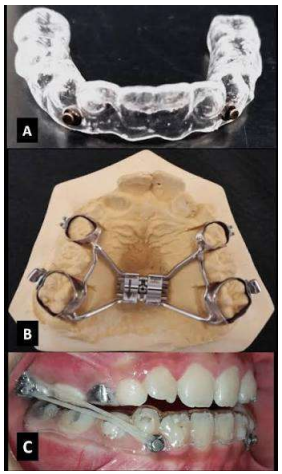




Metals and alloys

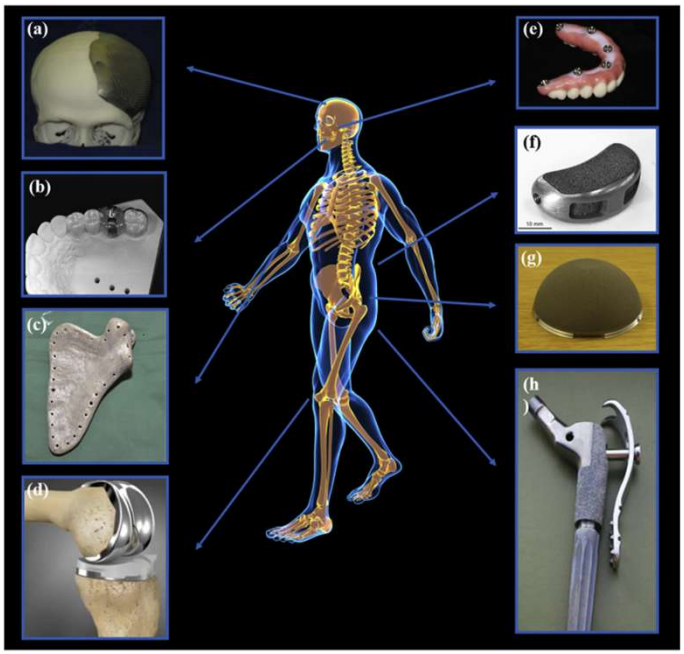
- Metals and alloys have a wide range of applications
  - Devices for fracture fixation
  - Partial and total joint replacement
  - External splints
  - Braces and traction apparatus
- Are susceptible to chemical and electrochemical degradation
- The implant materials may corrode and/or wear(metals and alloys)





Currently used metallic implant materials

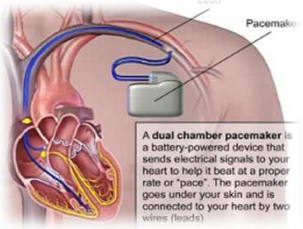
- Currently, approximately 70–80% of the clinically used implants are made of stainless steels, cobalt-chromium alloys, titanium alloys , nitinol , tantalum , and niobium.
- The fundamental criterion for choosing a metallic implant material is biocompatibility
- Metals and alloys used as implants must have required mechanical strength and reasonable corrosion resistance
- Metallic implants are made of;
  - Austenitic stainless steels
  - Cobalt–chromium alloys
  - Titanium and its alloys



Metals Used as Biomedical Materials:

- **Low-carbon alloy stainless steel**
(Features: Biocompatibility, Mechanical strength, Corrosion resistance)
Types: 316L, 316LVM
- **Cobalt-Chromium Alloys**
(Features: Biocompatibility, Wear resistance)
- **Titanium Alloys**
(Features: Biocompatibility, Mechanical strength, Corrosion resistance, Compatibility with medical imaging systems)
Types: Ti-6Al-4V, Ti-6Al-7Nb, Ti-13Nb-13Zr, Ti-12Mo-6Zr
- **Magnesium-based biodegradable metals**
- **Nickel-Titanium Alloys**
(Shape-memory smart metals)

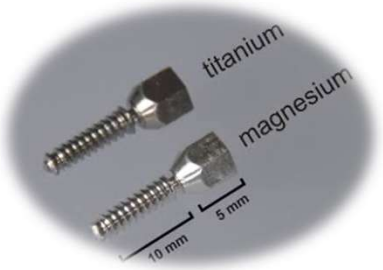








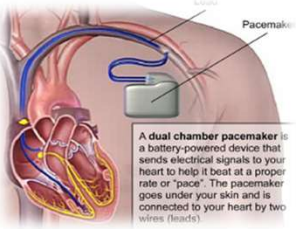




Metals Used as Biomedical Materials:

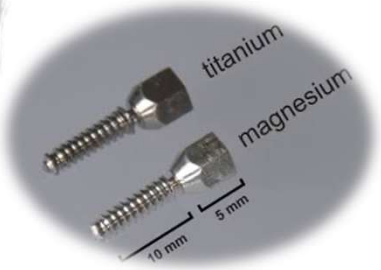
- **Tantalum**: Known for its high biocompatibility and corrosion resistance, commonly used in implants and bone repair applications.
- **Gold and Platinum**: Precious metals sometimes used in pacemaker electrodes and other devices where high conductivity and biocompatibility are critical.
- **Zirconium Alloys**: Known for corrosion resistance and biocompatibility, often alloyed with niobium (Zr-Nb) for implants.









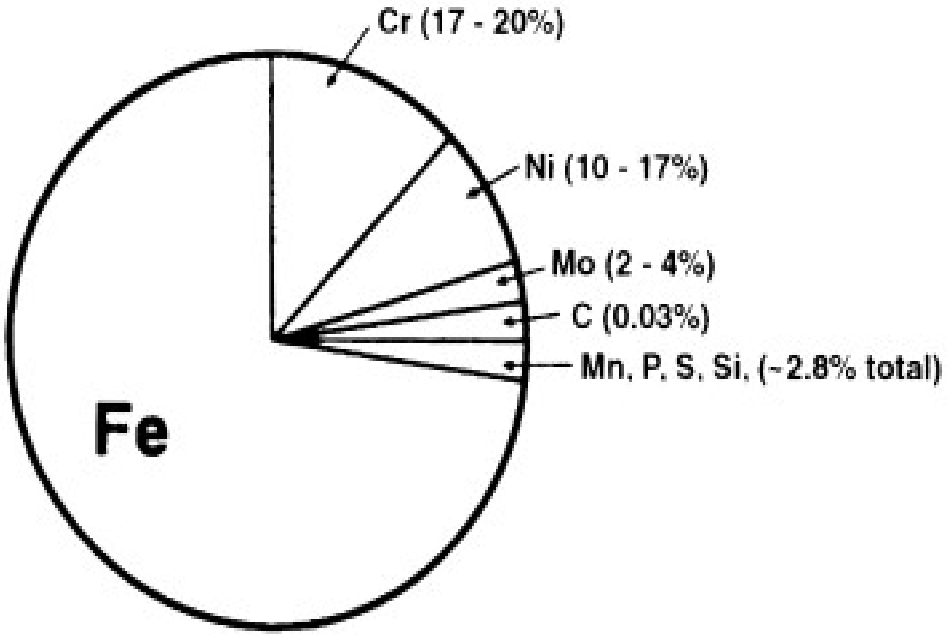


Stainless steel (SS)

- An alloy of **iron** and **carbon**.
- The form commonly used is **316L** (**3**% molybdenum; **16**% nickel & **L**= low carbon)
  - Other major alloying element include chromium
  - With minor amounts of
    - manganese, phosphorous, sulfur, and silicon.
    - ☻ Carbon provides strength and chromium provides the stainless quality to the stainless steel.
    - Nickle stabilizes the face-centered cubic austenitic phase
    - Chromium, Molybdenum, silicon - stabilizes the ferritic, body-centered cubic phase (weaker)

316L





Stainless Steel

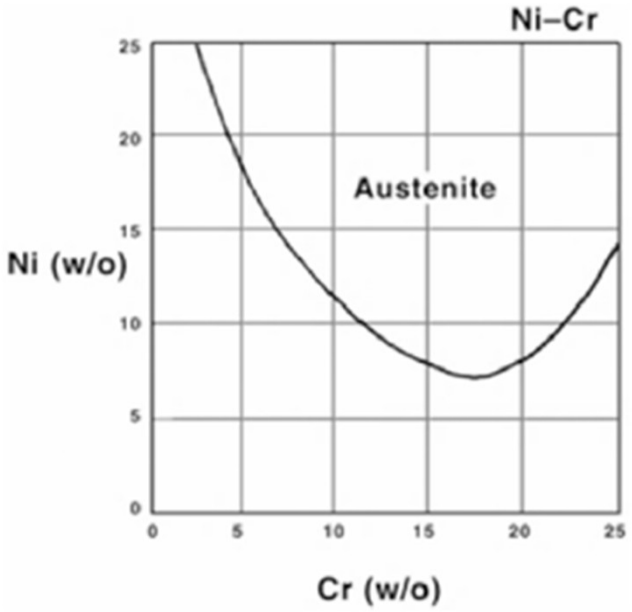
Stainless steel…

- **Chromium** also forms strongly adherent passive oxide film (Cr_2O_3 - corrosion resistance)
    - Nitric acid bath
        - Standardized method to enhance the passive oxide layer
        - limits the rate of electrochemical corrosion by about a **thousand to a million** times.
- **Carbon**
    - Must be kept at low
    - If high > **carbides** > chromium is used up > prone to corrosion related fractures
- Grain size
    - **100 microns**
    - Controlled by- Solidification process

Metallic Biomaterials

Functions of Cr and Ni in 316L

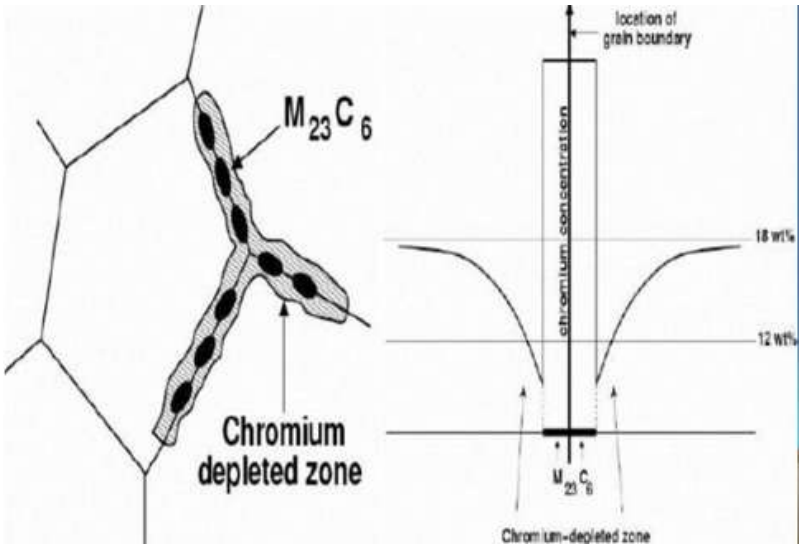
- The key function of chromium is to permit the development of corrosion-resistant steel by forming a strongly adherent surface oxide (Cr_2O_3).
- However, the downside to adding Cr is that it tends to stabilize the ferritic (BCC, body-centered cubic) phase of iron and steel, which is weaker than the austenitic (FCC, face-centered cubic) phase.
- Moreover, molybdenum and silicon are also ferrite stabilizers. So to counter this tendency to form weaker ferrite, nickel is added to stabilize the stronger austenitic phase.



Metallic Biomaterials

If C content is high in 316L !

- The main reason for the low carbon content in 316L is to improve corrosion resistance.
- If the carbon content of the steel significantly exceeds 0.03%, there is increased danger of formation of carbide, $Cr_{23}C_6$. Such carbides have the bad habit of tending to precipitate at grain boundaries when the carbon concentration and thermal history are favorable to the kinetics of carbide growth.



- The negative effect of carbide precipitation is that it depletes the adjacent grain boundary regions of chromium, which in turn has the effect of diminishing formation of the protective, chromium-based oxide Cr_2O_3.
- Steels in which such grain-boundary carbides have formed are called “sensitized” and are prone to fail through corrosion assisted fractures that originate at the sensitized (weakened) grain boundaries.

Stainless steel…

- **Advantages**
  - Very strong; biocompatible and reasonable corrosion resistance
  - Fracture resistant; relatively cheap and ductile
  - Easy to Process
- **Disadvantages**
  - Susceptible to corrosion(the most susceptible material to both crevice and galvanic corrosion)
  - Poor wear resistance
  - Stress shielding of bone due to superior stiffness (High Young's modulus- 200 G Pascals = 10x that of bone)
- Used in plates; screws; IMN; external fixators



Metallic Biomaterials

Concerns related to processing of stainless steel

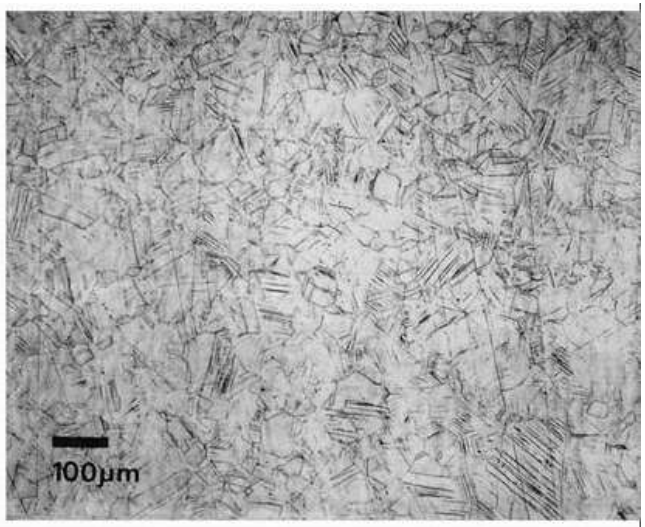
- The austenitic stainless steels **work-harden very rapidly**, and therefore cannot be cold-worked without intermediate heat-treatments.
- The heat-treatments, however, should not induce **formation of chromium carbide** (CCr_4) in grain boundaries, which might deplete Cr and C in the grains, causing corrosion. For the same reason, austenitic stainless steel implants are not usually welded.
- Another undesirable effect of heat-treatment is the **formation of surface oxide scales** that have to be removed either chemically (acid) or mechanically (sandblasting). After the scales are removed the surface of the component is polished to a mirror or matte finish. The surface is then cleaned, degreased, and passivated in nitric acid (ASTM Standard F86). The component is washed and cleaned again before packaging and sterilizing.

Microstructue of stainless steel

- ASTM standard (American Society for Testing Materials)
- Single phase: Austenite
- No ferrite no carbide present
- No sulfide (P, S maximum (<0,03)): İnclusions decreases corrosion resistance

Grain size in SS

- 316L: Low carbon stainless steel (L stands for Low)
- ≤ 100 μm
- **$N = 2^{(n-1)}$ (N:** number of grains in **$1inch^2$ at 100x magnification, n: ASTM** grain size**)**
- **Smaller grain size : higher strength**
- **ASTM grain size should be higher than 6.**
- Material should be homogeneous



Typical microstructure of cold-worked 316L stainless steel, ASTM F138,

Grain size in SS

- Grain size distribution should be uniform
- Smaller grains increases yield strength
- Hall-Petch equation

$$\sigma_y = \sigma_0 + \frac{k_y}{\sqrt{d}}$$

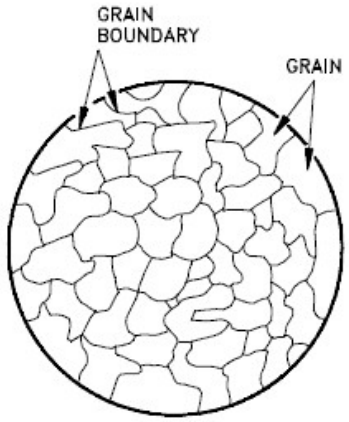
d: grain diameter

qy: yield strength

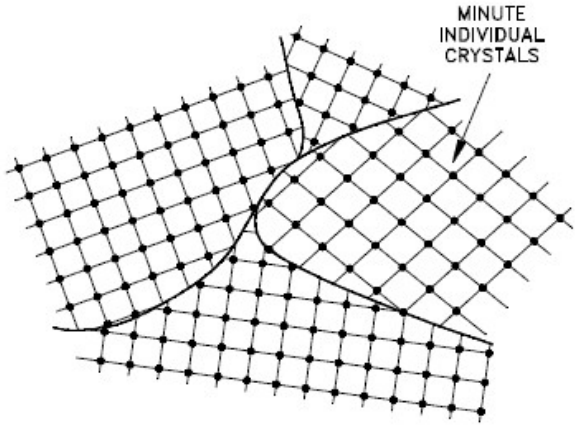
q0: frictional stess

k is a measure of the local stress needed to initiate plastic flow at a grain boundary

- Recrystallization
- Plastic deformation
- Annealing
- Solidification
- Generally applied method: 30% plastic deformation: High fatigue strength, Tensile strength,



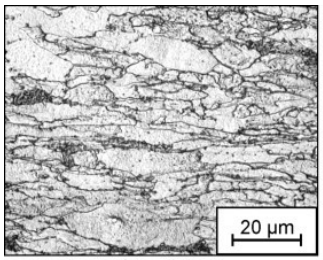
(A)



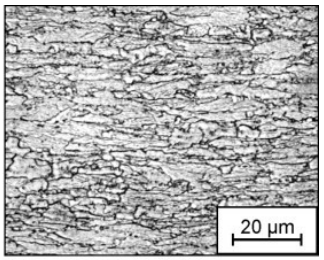
(B)

Plastic deformation: cold forging

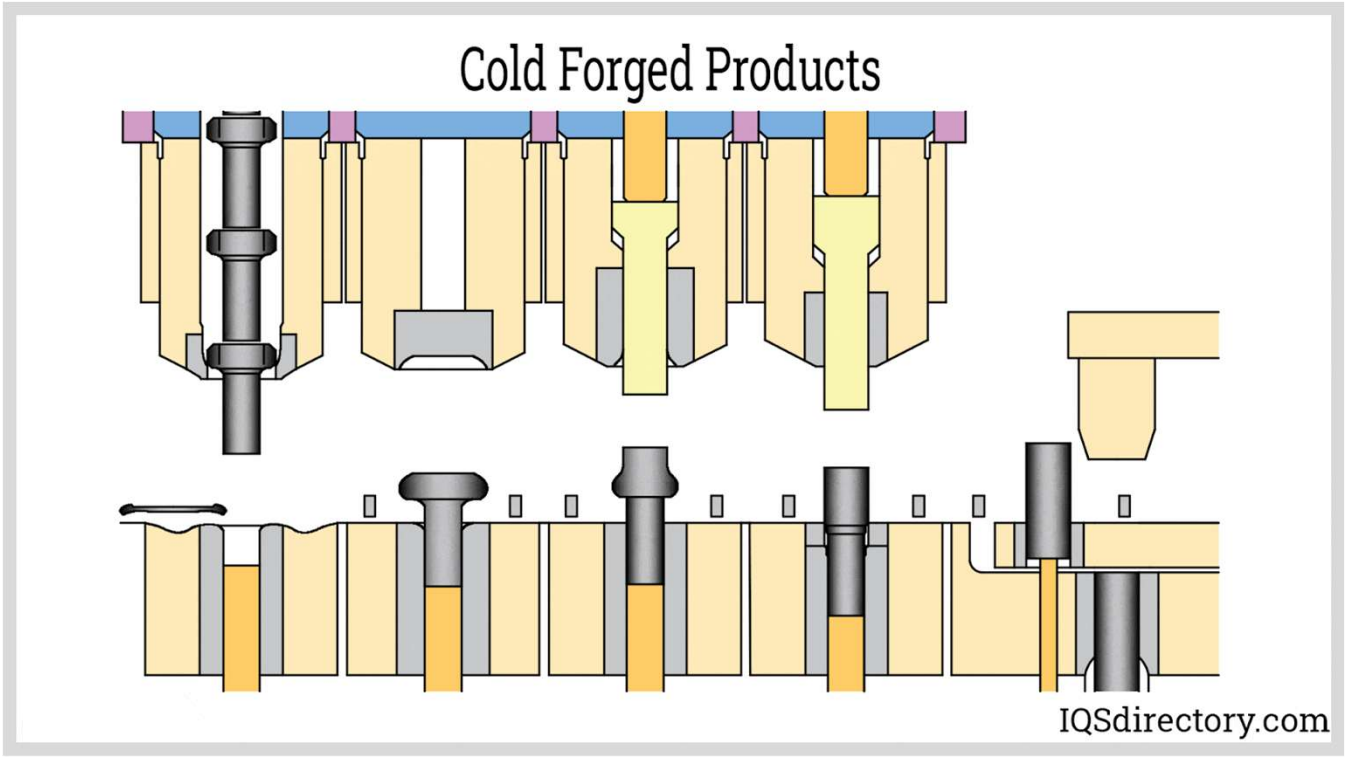
During cold forging grains elongate towards the forging direction (Texture evolution).



b) $\varepsilon = 30\ \%$



c) $\varepsilon = 75\ \%$



Mechanical properties of SS

Material	Content	Treatment	Elasticity modulus(GPa)	Yield strength (MPa)	Tensile strength (MPa)	Fatigue strength (MPa)
Stainless steel	S:~0,03 (high) P: ~0,045 (high) C: ~0,06 (high)	Annealed	190	221	483	221-280
Stainless steel	S: 0,01-0,03 (low) P: ~0,03 (low) C: ~0,03 (low)	Annealed	190	331	586	241-276
Stainless steel	S: 0,01-0,03 (low) P: ~0,03 (low) C: ~0,03 (low)	%30 cold forge	190	792	930	310-448
Stainless steel	S: 0,01-0,03 (low) P: ~0,03 (low) C: ~0,03 (low)	Cold forge	190	1213	1351	820

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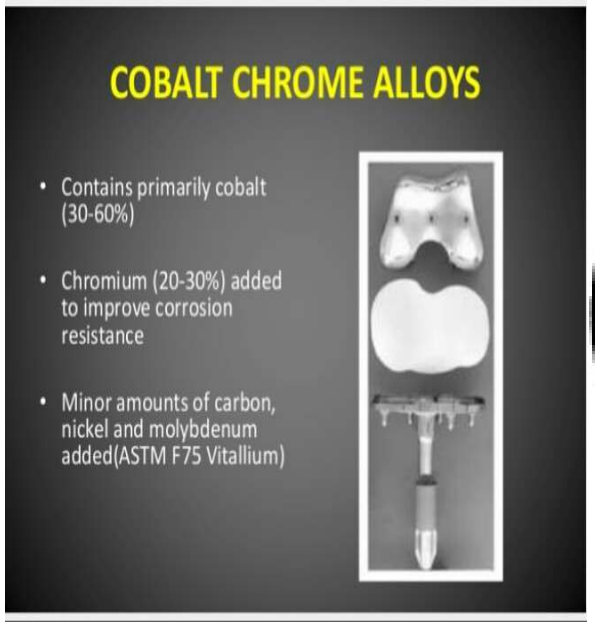
Cobalt-chromium alloys

- Primarily alloy of **cobalt** with **chromium**.
- The chromium forms a **strongly adherent oxide film** that provides a passive layer shielding the bulk material from the environment for corrosion resistance.
- Chromium added for corrosion resistance. (as with stainless steel)

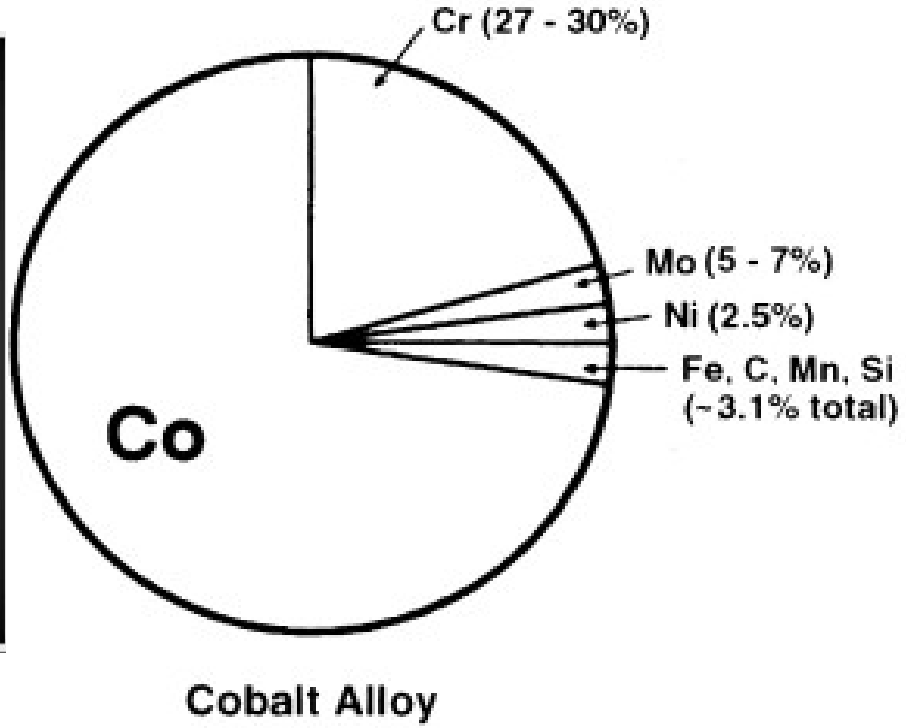




Cobalt-chromium alloys…







Cobalt Alloy

Metallic Biomaterials- Co-Cr alloy

1. **CoCrMo alloy:** Which is usually used to cast a product. It has been in use for many decades in dentistry and in making artificial joints.

2. **CoNiCrMo alloy**: Which is usually wrought by (hot) forging. It has been used for making the stems of prostheses for heavily loaded joints (such as the knee and hip).

The superior fatigue and ultimate tensile strength of the wrought CoNiCrMo alloy make it very suitable for applications that require a long service life without fracture or stress fatigue. Such is the case for the stems of the hip joint prostheses. This advantage is more appreciated when the implant has to be replaced with another one since it is quite difficult to remove the failed piece of implant embedded deep in the femoral medullary canal. Furthermore, the revision arthroplasty is usually inferior to the original in terms of its function due to poorer fixation of the implant.

Molybdenum is added to produce finer grains, which results in higher strength after casting or forging.

Cobalt-chromium alloys…

- Advantages
  - very strong
  - Ease of fabrication
  - better resistance to corrosion than stainless steel
  - excellent long term biocompatibility

Disadvantages

- risk of stress sheilding (**very high** Young's modulus)
- **expensive**

Uses:- usually for bearing surfaces; metal on metal devices eg. THR

Metallic Biomaterials- Co-Cr alloy

Processing of CoCr alloy

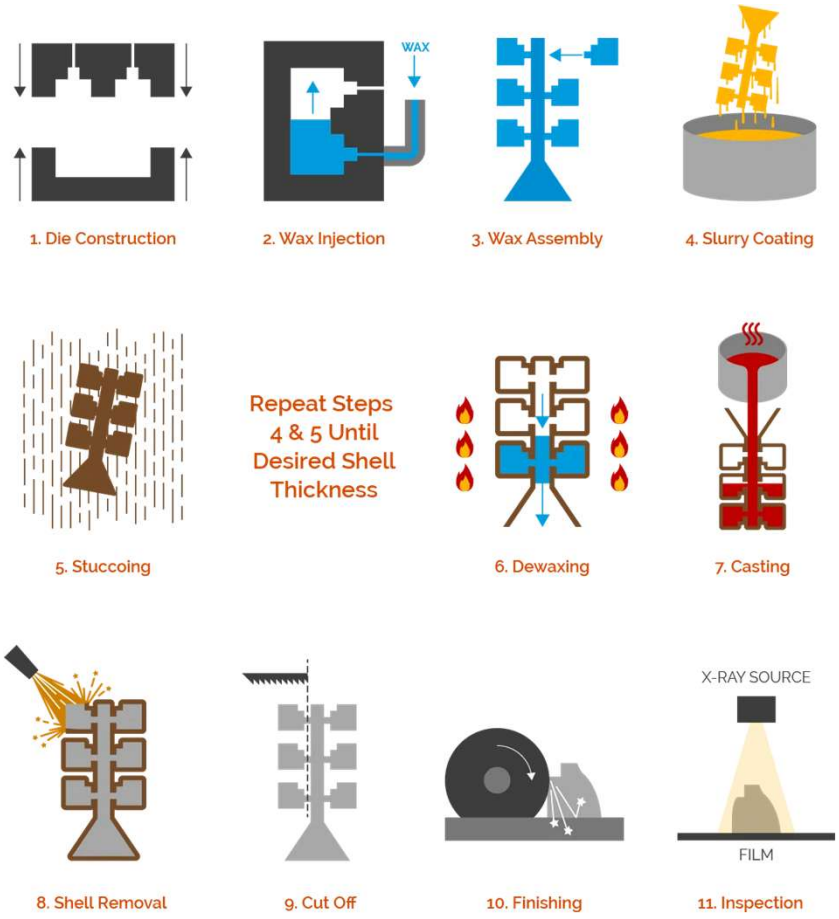
- The CoCrMo alloy is particularly susceptible to the work-hardening so that the normal fabrication procedure used with other metals cannot be employed.
- Instead the alloy is cast by a lost wax (or investment casting) method that involves the following steps

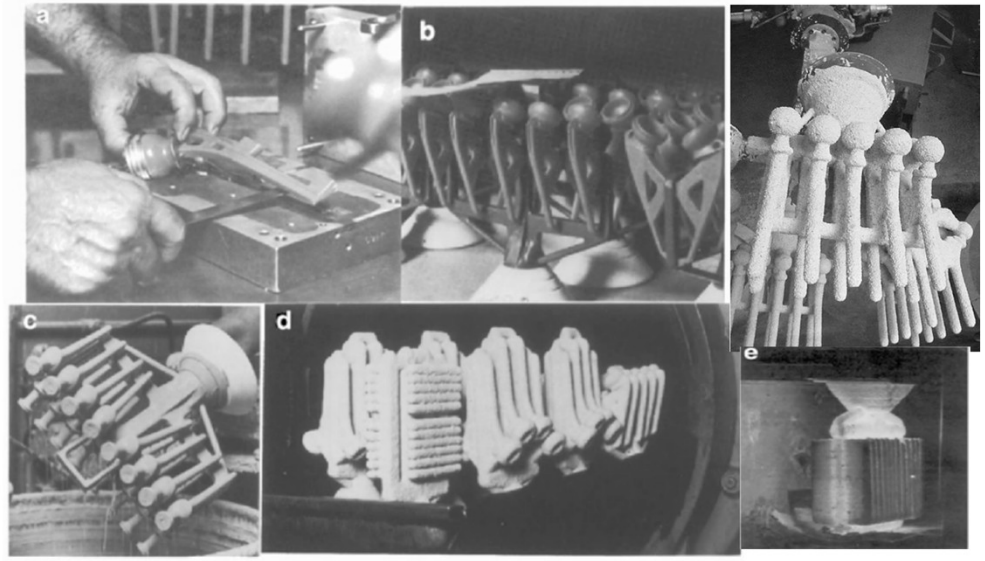
Investment casting-process

- A wax pattern of the desired component is made.
- The pattern is coated with a refractory material, first by a thin coating with a slurry (suspension of silica in ethyl silicate solution) followed by complete investing after drying.
- The wax is melted out in a furnace (100–150°C).
- The mold is heated to a high temperature, burning out any traces of wax or gas-forming materials.
- Molten alloy is poured with gravitational or centrifugal force. The mold temperature is about 800–1000°C and the alloy is at 1350–1400°C.

Metallic Biomaterials- Co-Cr alloy

Processing of CoCr alloy- Investment casting



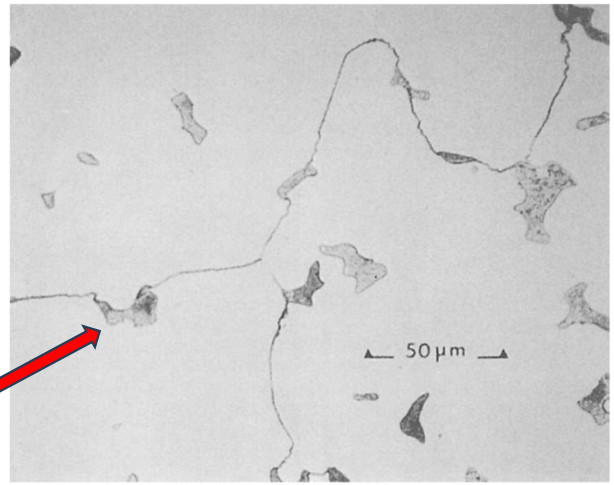


Lost wax casting of femoral joint prosthesis. (a) Injection of wax into a brass mold. (b) Wax patterns assembled for a ceramic coating (note the hollow part of the femoral head). (c) Application of ceramic coating. (d) A hot pressure chamber retrieves the wax, leaving behind a ceramic coating. (e) Pouring molten metals into the preheated ceramic mold.

Metallic Biomaterials- Co-Cr alloy

Concerns related to processing of Co-Cr alloy

- Controlling the mold temperature will have an effect on the grain size of the final cast; **coarse ones** are formed at higher temperatures. This is generally undesirable because it decreases the yield strength via a Hall–Petch relationship between yield strength and grain diameter. **Solution- Powder Metallurgy**

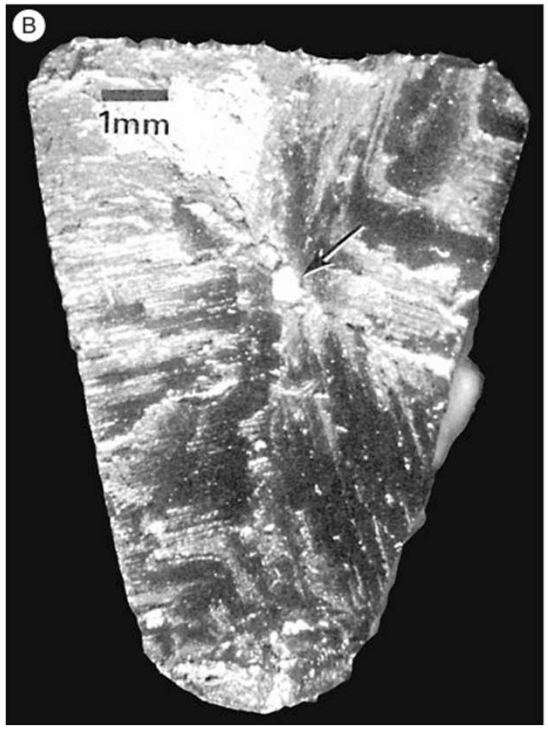


- Cast microstructure may consists of grain boundary **carbides ($M_{23}C_6$)** at high processing temperature. **Solution- Controlled Temp**
- Because of non-equilibrium cooling, a **"cored" microstructure** can develop. In this situation, the interdendritic regions become solute (Cr, Mo, C) rich, while the dendrites become depleted in Cr and richer in Co. This is an unfavorable electro chemical situation, with the Cr-depleted regions being anodic with respect to the rest of the microstructure. **Solution- Subsequent solution anneal heat treatments at 1225°C for 1 hour can help alleviate this situation.**

Metallic Biomaterials- Co-Cr alloy

Concerns related to processing of Co-Cr alloy

- Casting defects may arise. Figure shows an inclusion in the middle of a femoral hip stem. The **inclusion** was a particle of the ceramic mold (investment) material, which presumably broke off and became entrapped within the interior of the mold while the metal was solidifying. This contributed to a **fatigue fracture** of the implant device in vivo, most likely because of **stress concentrations** and **crack initiation** sites associated with the ceramic inclusion.



- For similar reasons, it is also desirable to avoid **macro- and microporosity** arising from metal shrinkage upon solidification of castings.

- Solution- **Powder metallurgy**

Metallic Biomaterials- Co-Cr alloy

Concerns related to processing of Co-Cr alloy

- To avoid problems such as the above with cast alloy, and to improve the alloy's microstructure and mechanical properties, powder metallurgical techniques have been used. For example, in hot isostatic pressing (HIP), a fine powder of the alloy is compacted and sintered together under appropriate pressure and temperature conditions (about 100 MPa at 1100◦C for 1 hour) and then forged to final shape. The typical microstructure (Fig. 8) shows a much smaller grain size (~8 μm) than the as-cast material.

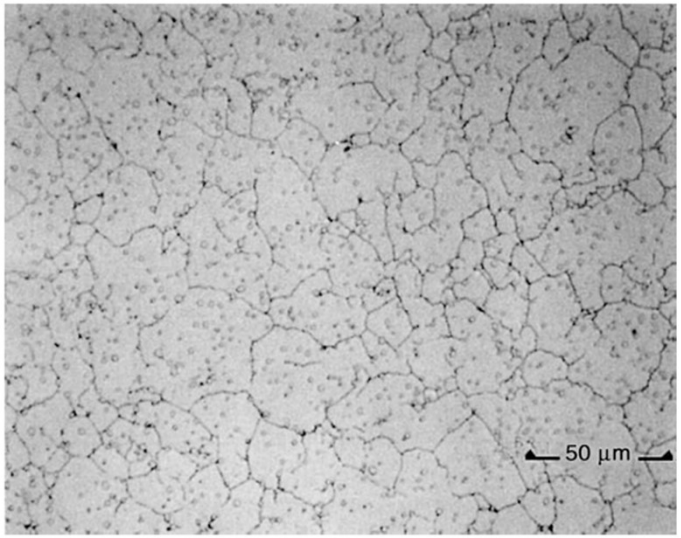


FIG. 8. Microstructure of the Co–Cr–Mo ASTM F75 alloy made via hot isostatic pressing (HIP), showing the much smaller grain size relative to that in Fig. 6. (Photo courtesy of Zimmer USA, Warsaw, IN.)

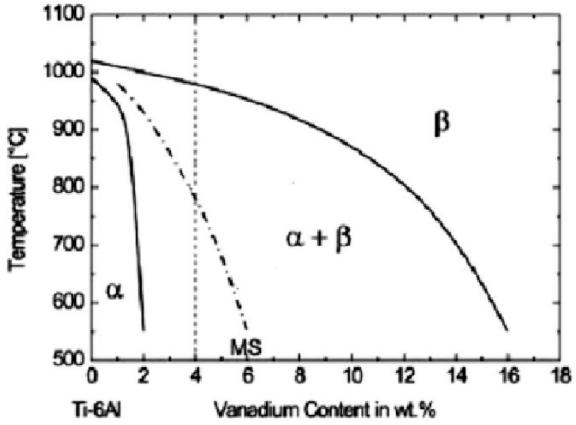
- Again, according to a Hall–Petch relationship, this microstructure gives the alloy higher yield strength and better ultimate and fatigue properties than the as-cast alloy. Generally speaking, the improved properties of the HIP versus cast F75 result from both the **finer grain size and a finer distribution of carbides**, which has a hardening effect as well

Metallic Biomaterials-Ti alloy

Ti alloys are used in hip joints, bone screws, knee joints, bone plates, dental implants, surgical devices due to its resistance to attack by body fluids, low density, high strength and low modulus.

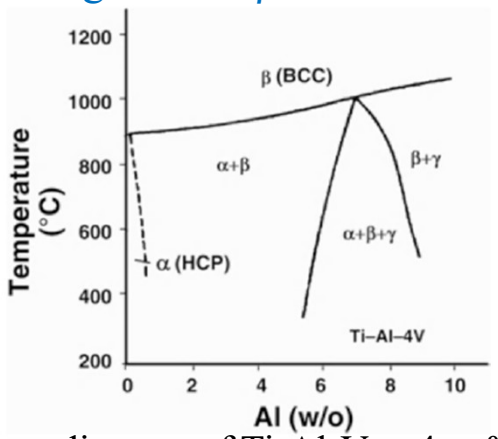
Commercially Pure (CP Ti) and Ti6Al4V alloy: Corrosion resistance comes from TiO_2. Titanium is an allotropic material that exists as HCP up to 882°C and BCC above that temperature. The addition of alloying elements to titanium enables it to have a wide range of properties:

Aluminum stabilizes α phase (HCP), i.e increases the α-β transformation temp



Phase diagram of Ti-Al-V at 6 wt% Al

Vanadium stabilizes β phase (BCC), i.e lowering the α-β transformation temp



Phase diagram of Ti-Al-V at 4 wt% V

β stabilizers are highly effective in improving strength by heat treatment

Metallic Biomaterials-Ti alloy

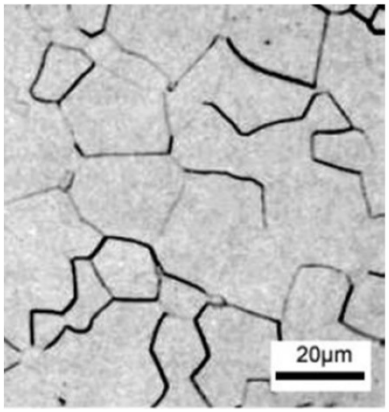
Concerns related to processing of Ti alloy

- Titanium is **very reactive** at high temperature and burns readily in the presence of oxygen. Solution- requires an inert atmosphere for high-temperature processing or is processed by vacuum melting
- Oxygen diffuses readily in titanium, and the dissolved oxygen embrittles the metal. Solution- hot-working or forging operation should be carried out below 925°C
- α alloys have single-phase microstructure, which promotes good weldability. The stabilizing effect of the high aluminum content of these groups of alloys makes for excellent strength characteristics and oxidation resistance at high temperature (300–600°C). These alloys cannot be heat-treated for strengthening since they are single phased since the precipitation of the second or third phase increases the strength by precipitation hardening process.

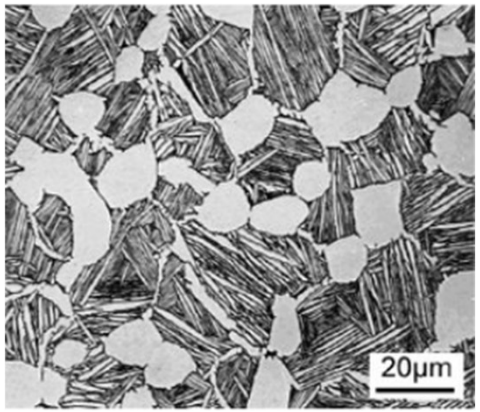
Metallic Biomaterials-Ti alloy

Concerns related to processing of Ti alloy

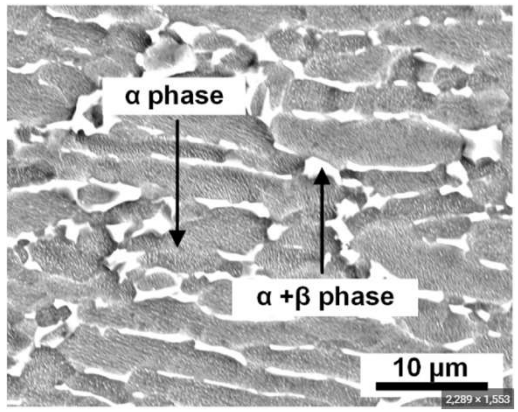
- The addition of controlled amounts of β-stabilizers causes the higher strength phase to persist below the transformation temperature, which results in the two-phase system. the precipitates of β phase will appear by heat-treatment in the solid solution temperature and subsequent quenching, followed by aging at a somewhat lower temperature.



Single phase



Dual phase



Dual phase

Titanium and Titanium Alloys

- Minimal attenuation problem on MRI
- High strength to weight ratio
  - Density of 4.5 g/cm3 compared to 7.9 g/cm3 for 316 SS
- Modulus of elasticity for alloys is about 110 GPa
  - Not as strong as stainless steel or cobalt based alloys, but has a higher "specific strength" or strength per density
  - Low modulus of elasticity

Titanium Alloys

- Primary alloying elements are **aluminum** and **vanadium.**
- Adherent passive layer of **titanium oxide** (TiO_2) significantly exceeds the other alloys (stainless steel and the cobalt alloys) in corrosion resistance.
- Excellent pitting; intergranular and crevice corrosion resistance
- Uses: fracture plates (esp. compression plates), screws, intramedullary nails, some femoral stems.

Titanium Alloys

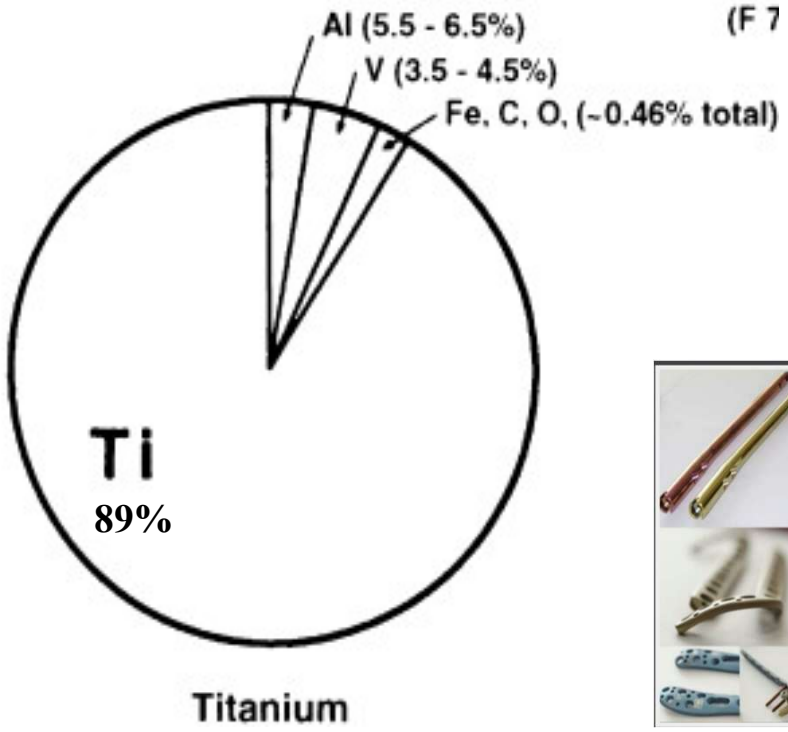
- Co-Ni-Cr-Mo-Ti, Ti6A4V
- Poor shear strength which makes it undesirable for bone screws or plates
- Tends to seize when in sliding contact with itself or other metals
- Poor surface wear properties - may be improved with surface treatments such as nitriding and oxidizing

Titanium Alloys…

The most common titanium alloy used in orthopedic surgery is **T64** (Ti-6A1-4V).

☼Osseointegration

Titanium has been shown to promote good bone apposition to its surface when it is implanted, and porous surfaces have proven to be receptive to bone ingrowth.





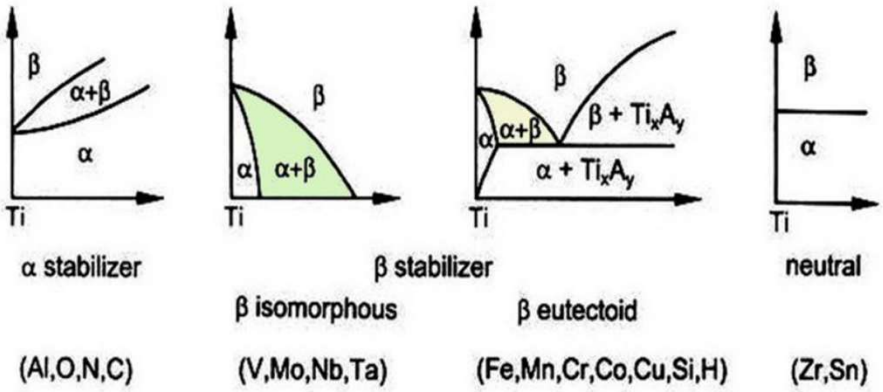
Alloying elements

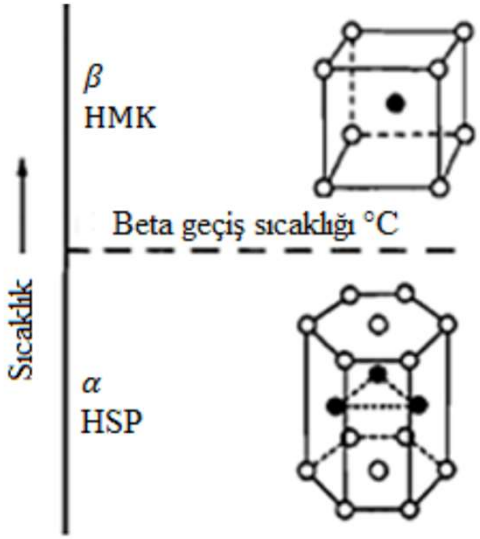
- Pure titanium generally does not have high yield or fatigue strength due to its ease of dislocation movement, but these properties can be significantly enhanced through alloying.

- In titanium alloys, adding elements like aluminum (Al) and vanadium (V) helps impede dislocation movement and enhances both yield strength and fatigue resistance.

- Ti6Al4V: Dual phase alloy is the most common in Biomedical research
- Al → alpha stabilizer (HCP)
- V → Beta stabilizer (BCC)

-





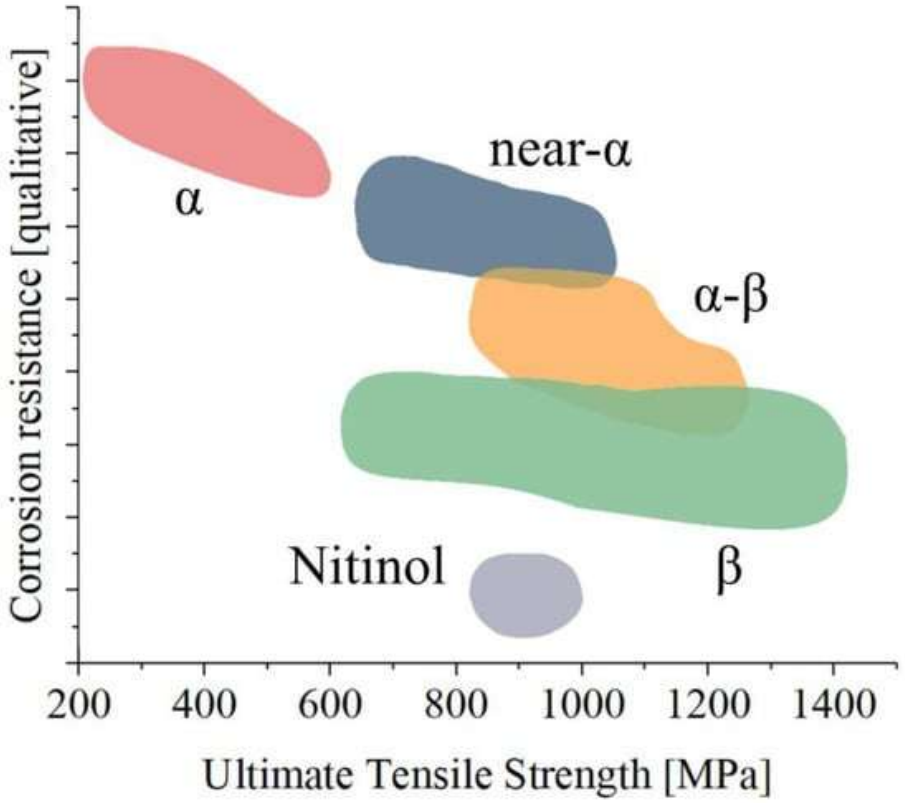
Ti6Al4V

Ti-6Al-4V (dual phase), has a combination of strength, toughness, and formability.

- Beta titanium alloys are not as widely used in biomedical applications due to Lower biocompatibility of Alloying Elements: Many beta titanium alloys contain elements like vanadium, aluminum, and niobium. While Ti-6Al-4V (which contains vanadium and aluminum) is commonly used in biomedical applications, the long-term biocompatibility of some beta alloys, especially those with vanadium, is questioned due to potential toxicity. For example, vanadium can release ions that may be toxic or cause adverse reactions in the body.

Alpha-phase titanium alloys, characterized by a high proportion of alpha-phase crystals, are generally more resistant to corrosion due to their stable microstructure. In contrast, beta-phase titanium alloys, rich in beta-phase crystals, can be more prone to corrosion, especially in aggressive environments.

Ti6Al4V



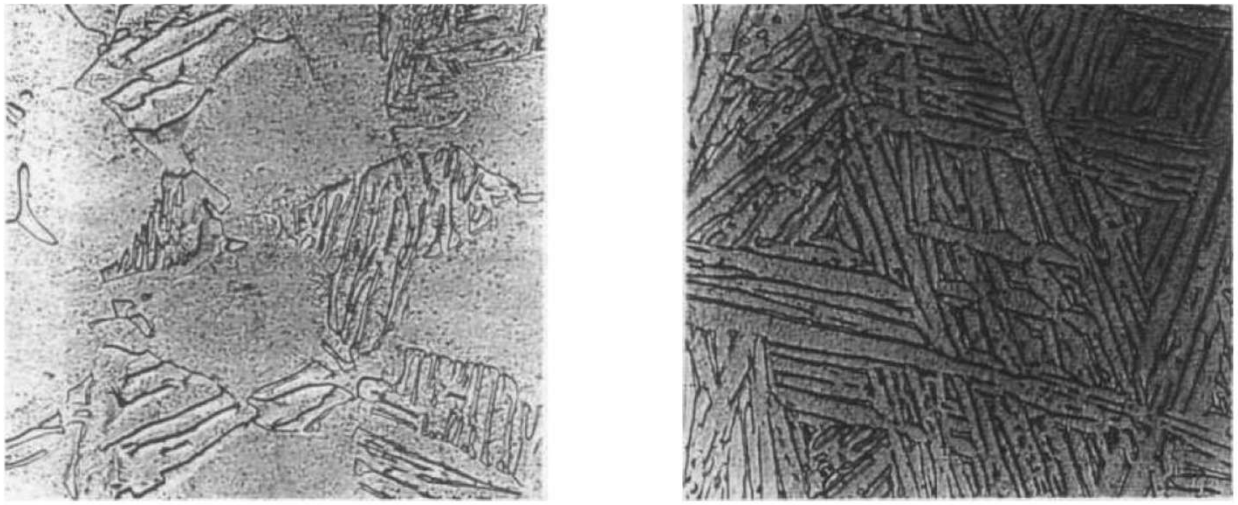
Relationship between the ultimate tensile strength and the qualitative corrosion resistance of titanium alloys applied in the biomedical field. The colored areas indicate the kind of alloys.

Heat Treatment and Phase Transformation at High Temperatures

- Holding at 1000°C: When Ti-6Al-4V is held at high temperatures (such as 1000°C) for an extended period, the alloy can enter a state where it is almost entirely in the beta phase. This is because 1000°C is above the beta transition temperature (approximately 995°C for Ti-6Al-4V), at which point the alloy transitions to a fully beta-phase structure.
- Slow Cooling: If the alloy is slowly cooled from this high-temperature beta phase, it forms a Widmanstätten structure. This structure consists of needle-like or plate-like alpha phase precipitates within the beta phase, which typically results from a slower cooling rate.

- **Widmanstätten Structure**: This microstructure is known for having a needle-like or plate-like alpha phase within the beta phase matrix: lower strength, reduced ductility.

Widmanstätten Pattern



(a) (b)

a) And b) needle-like α phase evolution (Widmanstatten)

The most common microstructure observed during cooling is the needle-like α structure transformed from the β structure. The α grains form by nucleating and growing within the β matrix on one or several planes. Figure 2.16 shows needle-like α (Widmanstätten) structure forming on one or multiple planes (Ribeiro et al., 2003).

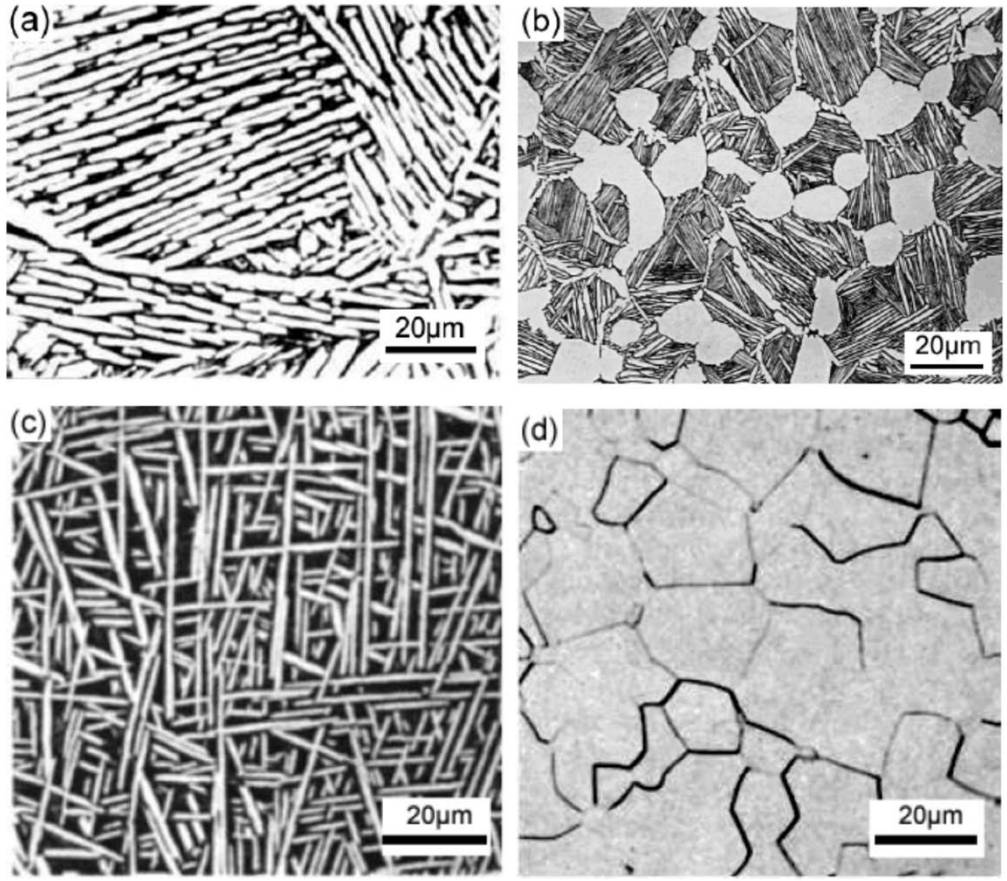
Heat treatment

*After cold forming, it is necessary to apply aging or stress-relieving processes, or to perform hot forming.

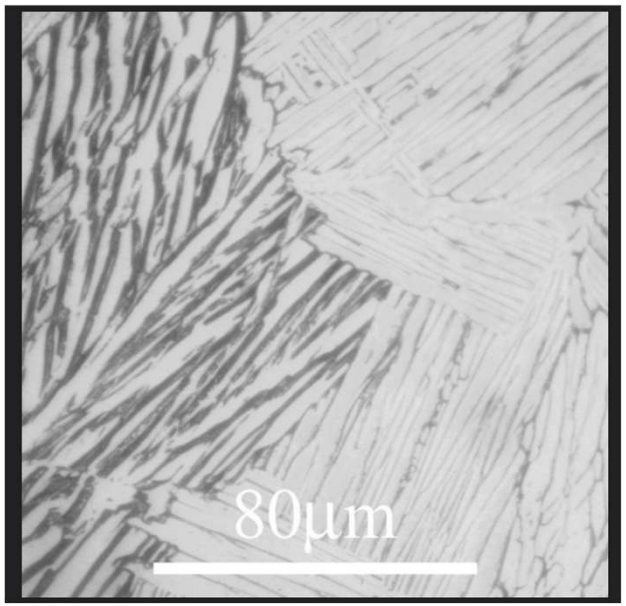
*Hot forming is generally carried out at temperatures ranging from 595 to 815 °C. This process facilitates easier deformation of the material and minimizes springback by relieving the stresses generated during deformation.

*Annealing and stress-relieving processes can be applied not only to enhance corrosion resistance and prevent warping, but also to prepare the material before shaping.

Ti-Al-V: Forged alloys



Typical microstructures of α+β titanium alloys: (a) Widmanstätten; (b) duplex microstructure; (c) basket-weave microstructure; (d) equiaxed structure (adapted from [54]).



This sample is a mixed α+β Ti alloy with a Widmanstätten microstructure. The α phase (h.c.p) precipitates out as Widmanstätten plates upon cooling from the β phase (b.c.c). The matrix (dark) appears as thin layers between the a plates (light). The crystallographic relationships are very precise in this example giving h.c.p α plates in a b.c.c β matrix.

Ti-Al-V: Forged alloy

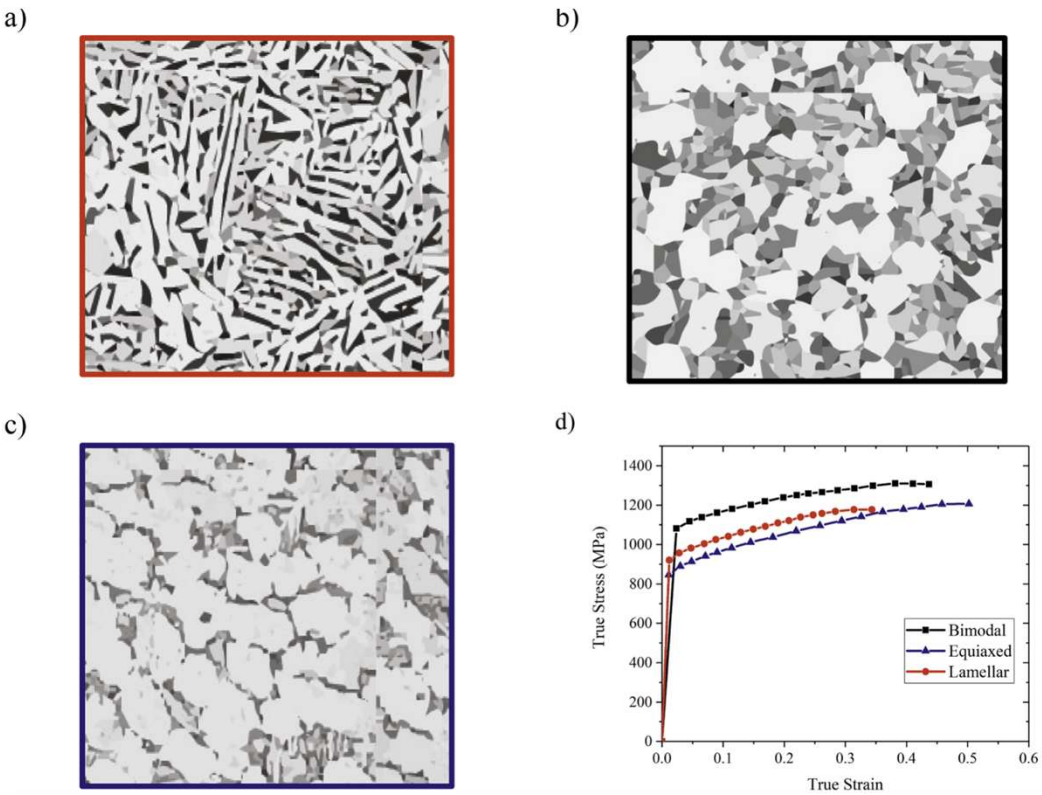
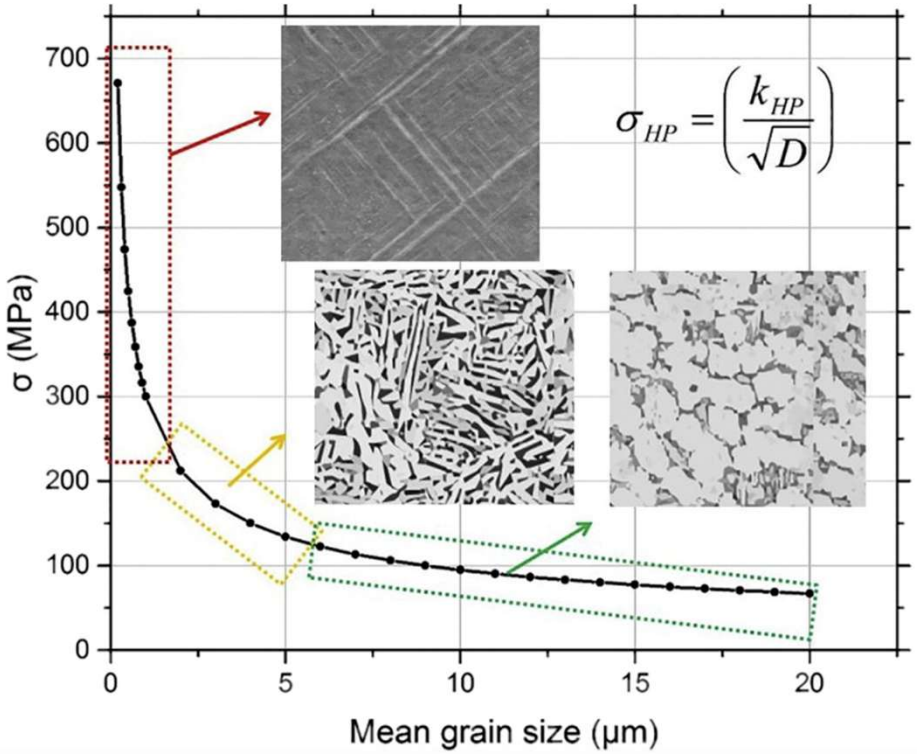


Fig. 1. Typical wrought Ti-64 microstructures: (a) lamellar (b) bimodal (c) equiaxed and (d) their representative mechanical response, at room temperature and a strain rate of 10^{-3} s^{-1}

Effect of grain size on strength



Grain boundary strengthening in Ti-64's σ_Y.

Ti-Al-V : Forged alloy

Table 2 Tensile properties of the tested cylindrical bars

Materials	Young's Modulus [Gpa]	Yield Strength [Mpa]	Ultimate tensile strength [Mpa]	Percentage of Elongation [%]	Percentage of reduction in Area [%]
Wrought Ti-6Al-4V f (Longitudinal)	105	948	994	21	40
Wrought Ti-6Al-4V (Transverse)	107	962	1008	19	38

Titanium Alloys…

- **Advantages**
  - very biocompatable
  - forms adherent oxide coating through self passivation (exceptional corrosion resistant)- form its passivation layer in air before implantation
  - Ductile and fatigue resistant
  - low modulus of elasticity makes it more similar to biologic materials as cortical bone.
  - MR scan compatability

Disadvantages

-poor resistance to wear (notch sensitivity) (not used as a femoral head prosthesis)

-generates more **metal debris** than cobalt chromium (Vanadium is cytotoxic).

-relatively expensive

Mechanical properties of titanium alloys

- No change in elastic modulus
- Alloying increases fatigue strength

Alaşım		Tavlama Isıl İşlem Parametreleri		
İç Yapı	**Türü**	**Sıcaklık *(°C)***	**Zaman *(Saat)***	**Soğutma Ortamı**
α	Saf titanyum	660-770	0,1-8	Havada
	Ti-5Al-2,5Sn	730-850	0,16-4	Havada
$\alpha+\beta$	Ti-4Al-4V	660-770	1-8	Havada / Fırında
	Ti-6Al-4V	715-850	1-8	Havada / Fırında
β	Ti-13V-11Cr-3Al	720-800	0,16-1	Havada / Suda

Most α+β alloys have high-strength and formability, and contain 4-6 wt% of β-stabilisers which allow substantial amounts of β to be retained on quenching from the β→α+β phase fields, *e.g.* Ti-6Al-4V. Al reduces density, stabilises and strengthens α while vanadium provides a greater amount of the **more ductile β phase** for hot-working. This alloy, which accounts for about half of all the titanium that is produced, is popular because of its strength (1100 MPa), creep resistance at 300°C , fatigue resistance and castability.

Microstructure of titanium

- CP titanium (commercial pruity)– Hexagonally closed packed
- Grain size 10-150 μm

Most titanium utilized for the production of biomedical devices is alloyed with other elements, and even the commercially pure titanium grades actually contain small amounts of trace elements that have a clear influence on their final properties.

Effect of intersticial atoms

- O- 0,1-0,4 %
- N - max 0,5 %
- C – 0,1 % (Generally oxygen and nitrogen is used to strenghten the material)

- O, N, C are intersticial atoms in titanium atoms (spaces between titanium atoms in the crystal structure), they induce solution hardening (distortion of the crystal lattice due to interstitial atoms, making dislocation movement more difficult.).
- Effect of nitrogen is double the other elements.

- The effect of increasing oxygen content from 0.18% to 0.4% on titanium alloys is indeed significant for yield strength and fatigue strength.
- The fatigue strength of titanium alloys can indeed increase from around 88 MPa to much higher values (e.g., up to 216 MPa in certain cases) with optimized oxygen content.

Effect of passive oxide layer

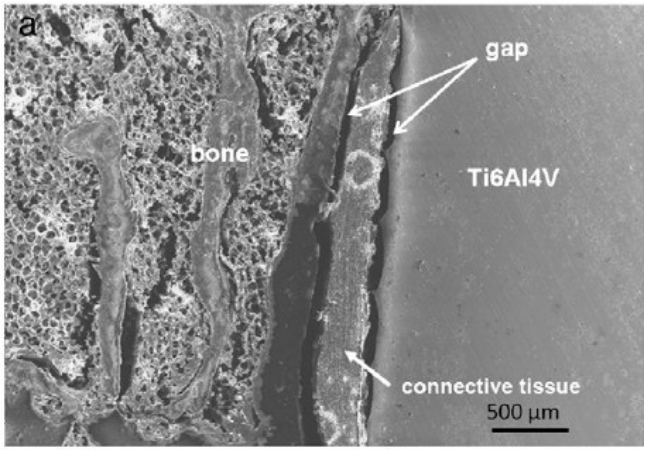
Titanium is a reactive material. When it is exposed to the environment, it reacts with oxygen and forms a protective oxide layer (passive oxide layer) that reduces the oxidation process.

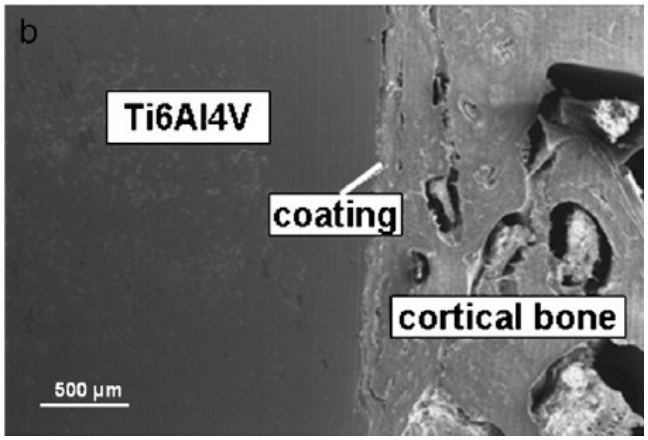
The passive layer consist mainly of TiO_2

This oxide layer, primarily composed of titanium dioxide (TiO_2), is stable, inert, and effectively isolates the underlying metal from the surrounding biological environment.

The oxide layer's surface chemistry fosters the adsorption of biomolecules like proteins

Surface modification of titanium alloys





(a) Bioinert materials response by formation of a non-adhering connective tissue (fibrous tissue) capsule separating an uncoated Ti6Al4V implant surface (right) from the surrounding bone (left) after implantation into the femoral condyle of an adult dog for 6 months. (b) Bioactive materials response characterized by tight adherence of a hydroxyapatite-coated Ti6Al4V implant (left) to the surrounding cortical bone (right) after implantation into the femoral condyle of an adult dog for 6 months

1. Hydroxyapatite (HA) Coatings
2. Calcium Phosphate Coatings

10.1016/j.surfcoat.2012.11.013

Applications of titanium alloys in BM research

Applications of Dual phase titaniums:
Dental implants
Joint replacements
Trauma devices
Spinal implants
Cardiovascular devices
Soft tissue implants



Applications of alpha Titaninum:

Dental Implants: CP-Ti is commonly used for dental implant screws and abutments due to its excellent biocompatibility and ability to integrate well with bone.

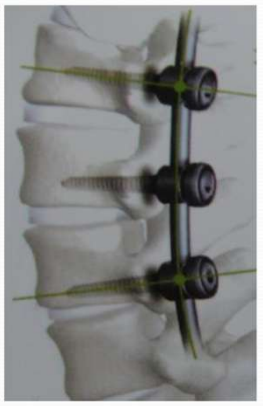
Orthopedic Implants: Although less common than Ti-6Al-4V, CP-Ti is sometimes used in non-load-bearing orthopedic implants due to its biocompatibility.

Pacemaker and Defibrillator Housings: CP-Ti's biocompatibility and corrosion resistance make it an ideal material for the cases of electronic medical devices implanted in the body.

Applications of titanium alloys in BM research

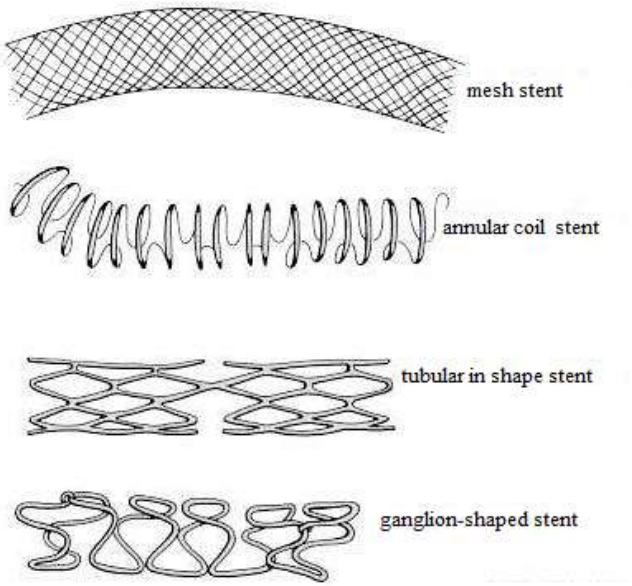
- Dental applications: For applications that require very high mechanical strength without compromising biocompatibility, such as in bone plates, screws, fasteners, abutments used for oral implants, and brackets for orthodontic appliances
- Ortheopedic applications: most widely used titanium alloy in orthopedics, is applied in various load-bearing applications, including hip and knee implants.
- Trauma devices, such as bone plates, screws, and intramedullary nails, require materials that can withstand the mechanical stresses imparted on them within the human body, which can be superior to those faced by orthopedic devices.
- Spinal implants: Titanium pedicle screws are used to stabilize and immobilize the spine: they are inserted into the vertebral pedicles and serve as anchors for other spinal implant components.





Applications of titanium alloys in BM research

- *4. Cardiovascular Devices:* Titanium and its alloys play a crucial role in the development of cardiovascular devices, contributing to improved patient outcomes in the treatment of various heart and vascular conditions.





Cardiovascular stents

Best Performance among the three

- Titanium has the best biocompatibility of the three.
  - Metal of choice where tissue or direct bone contact required (endosseous dental implants or porous un-cemented orthopedic implants)
  - Corrosion resistance due to formation of a solid oxide layer on surface (TiO2) -- leads to passivation of the material

COMPARISON OF METAL ALLOYS

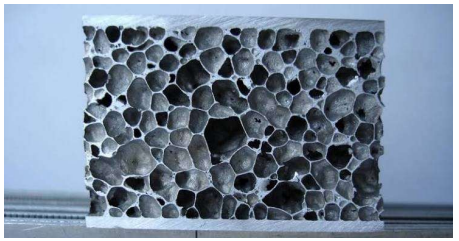
ALLOY	Young's modulus (GPa)	Yield strength (MPa)	Ultimate tensile strength (MPa)
Stainless Steel 316L	190	500	750
Titanium 64	110	800	900
Cobalt chrome F562	230	1000	1200

Metallic Foam

- Types of metallic foams
  - Solid metal foam is a generalized term for a material starting from a liquid-metal foam that was restricted morphology with closed, round cells.
  - Cellular metals:A metallic body in which a gaseous void is introduced.
  - Porous metal: Special type of cellular metal with certain types of voids, usually round in shape and isolated from each other.
  - Metal Sponges: A morphology of cellular metals with interconnected voids.



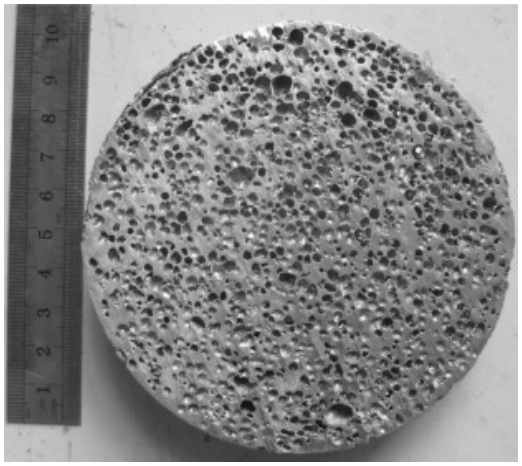




Magnesium Foam

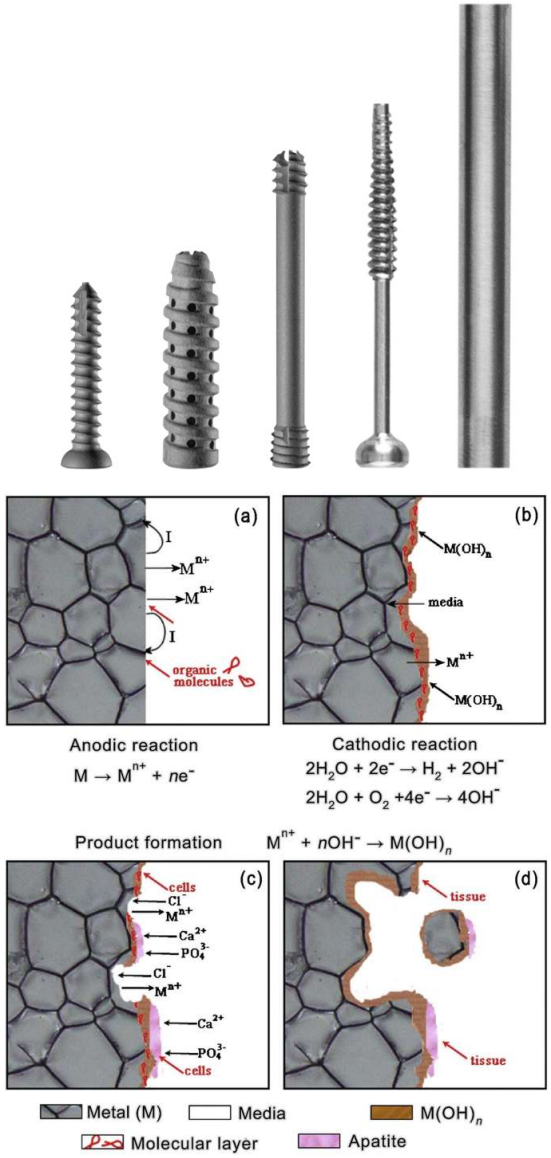
Open cellular structure permits ingrowths of new-bone tissue and transport of the body fluids

- Strength & Modulus can be adjusted through porosity to match natural bone properties



Why Magnesium?

- Bioresorbable
- Biocompatible
  - Osteoconductive
  - Osteoinductive
- Properties of bone can be easily attained using varying processing techniques



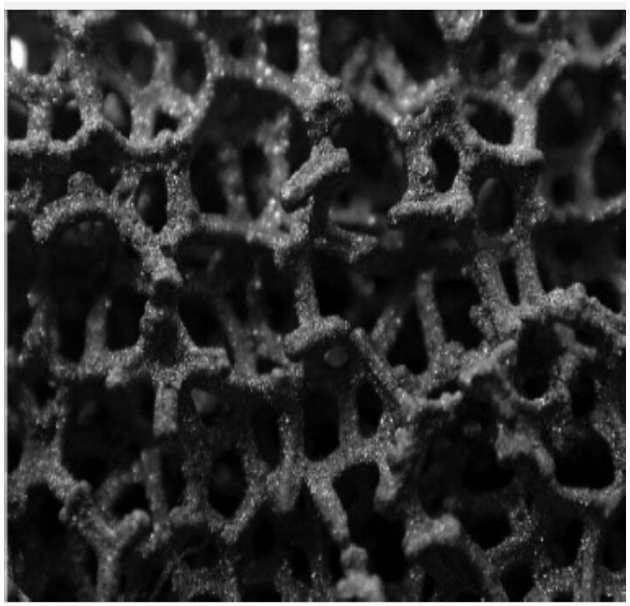
Comparisons

Material	Density	Youngs Modulus	Tensile Strength	Estimated Cost Ranking
Bone	2.3	10 – 40	3 – 20	Na
Stainless Steel	7.9	196	290	1
Co Alloys	8.9	211	345	4
Ti Alloys	4.5	105	200	3
Mg Foam	2.33	10.476	2.843	2

Tantalum

- The latest metal to hit the orthopedic market is tantalum.
- The benefit of tantalum is the ability to form it into **porous foams** with a structure on the order of trabecular bone, providing a **scaffold** that is optimum for bone ingrowth.
- The mechanical properties of this novel metal depend on its porosity and structure but are sufficient to provide mechanical support during the **period of bony integration**
- Tantalum has been shown to be virtually **inert**, provoking a minimal tissue response.
- This combination of properties has lead to the development of tantalum structures for the **backing of acetabular cups** and **spinal fusion** cages.
- It shows great promise for **future!**

Tantalum…





Left- porous tantalum micro-stracture
Right -Top row: monoblock acetabulum and acetabular augment
- Middle row: tibia; TKA augment; salvage patella button
- Bottom row: osteonecrosis and spine arthrodesis implants

Tantalum…

TABLE I Mechanical Properties of Commonly Used Alloys

Property	Porous Tantalum	Ti-6Al-4V	Co-Cr-Mo	316L Stainless Steel
Modulus of elasticity *(GPa)*	2.5-3.9	106-115	210	230
Yield strength *(MPa)*	35-51	860	825	170-690
Ultimate strength *(MPa)*	50-110	780-1050	430-1028	515-860
Hardness (on Rockwell C scale)		36	42	60
Fatigue strength *(MPa at 10^7 cycles)*	18-20	480-590	310	180-300
Mass density *(g/cm^3)*		4.4	8.3	8.0

Detrimental factors affecting the performance of implant materials

1.Mechanical forces imposed on the implant

- The complicated service conditions and loads encountered on devices implanted in the human body are generally quite high.
- Implants are subjected to both static and dynamic loads

2.Biological environment

- The human body is a harsh environment for metals and alloys (oxygenated saline solution 0.9%, pH 7.4, and temperature-37+/-1 C)
- Implantable materials undergo chemical/electrochemical dissolution

Detrimental factors…….

3.Tissue–implant corrosion

- The interactions between the material and the tissues is of prime importance
- Such interactions induce corrosion/ionization of the implanted device
- Corrosion can have two effects on the implant
  - May weaken and the premature failure will result
  - Release of corrosion products from the implant

- No metallic material is totally resistant to corrosion or ionization within living tissues

Corrosion in metals

- Crack formation and fatigue: This is a fundamental mechanism of metal degradation.
- Corrosion in the body is Metal atoms ionize, transfer into solution and combine with oxygen (Biological fluids contain oxygen).

- Corrosion has a relation with fatigue.

- Increasing amount of alloying element may have a deleterious effect on implant. Ions may relase and accumulate in certain locations of the body.

Surgeons must avoid damaging the passive layer on the metal surface when placing it into the body. This passive layer, a thin oxide coating, is essential for protecting the metal against corrosion.

Detrimental factors……

Other variables

1. **Direct overloading;** If excess load is transmitted to the implant fixed on the fractured bone slit

2. Fatigue loading: Fatigue failure is defined as failure due to dynamic as well as cyclic loading of stresses imposed on the implant

3. Wear: mechanical removal of materials during the process of relative motion between surfaces

- Four types of mechanisms may be implicated in the wear of metal components of joint prostheses

(i) abrasive, (ii) adhesive, (iii) fatigue and (iv) corrosive modes.

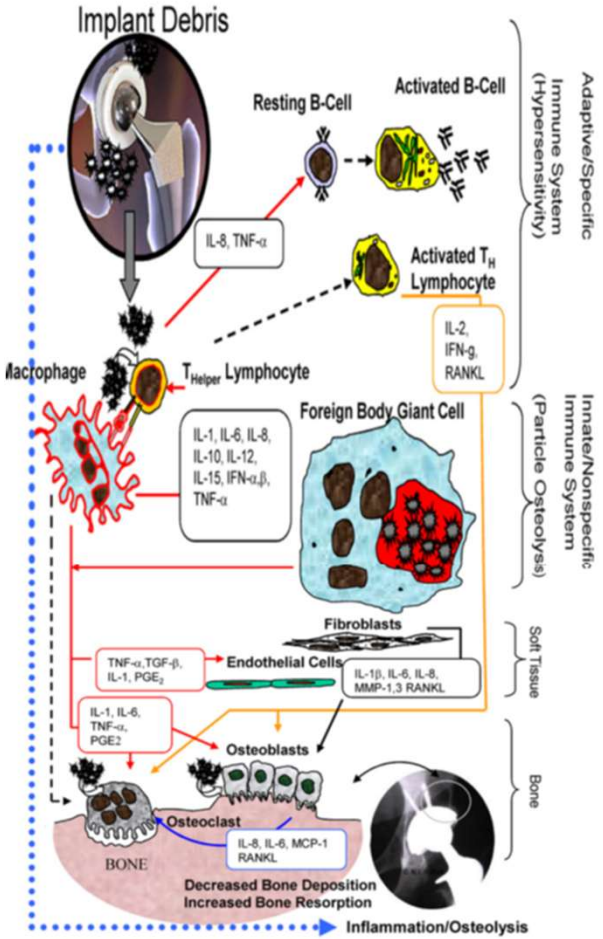
Failure of bioimplants

- Most of bioimplants fail after 12-15 years
- The reasons for their failure includes
  - Mechanical,
  - Chemical,
  - Tribological,
  - Surgical,
  - Manufacturing and biocompatibility issues

Corrosion has remained as one of the challenging clinical problems

Inflammation and bioimplants….

- Local events takes place following implantation of biomaterials
  - Injury
  - Blood-material interactions
  - Provisional matrix formation
  - Acute inflammation
  - Chronic inflammation
  - Granulation tissue development
  - Foreign-body reaction (FBR)
  - Fibrosis (fibrous capsule development)



Inflammation and bioimplants….

- Inflammation is characterized by
  - The presence of mononuclear cells (monocytes and lymphocytes) at the implant site
  - Proliferation of blood vessels and connective tissue

- The persistence acute and/or inflammatory responses beyond three-week period indicates an infection

Microbial Colonisation of Biomaterials

- Biomaterials are not inert
- Their surfaces are capable of physical, chemical and biological interactions that facilitate their colonisation by microbes
- Soon after implantation, all biomaterials become coated with glycoproteins from body fluids
- These films are formed by proteins (fibronectin, fibrinogen, vitronectin, laminin, collagen)
- They then provide receptor sites for the attachment of bacteria.

Microbial Colonisation of Biomaterials...

Many bacteria and fungi spp. are involved in biomaterial infections, but most belong to;

- Endogenous bacteria or
- The commensal community of the skin such as Staphylococcus epidermidis and Staphylococcus aureus.

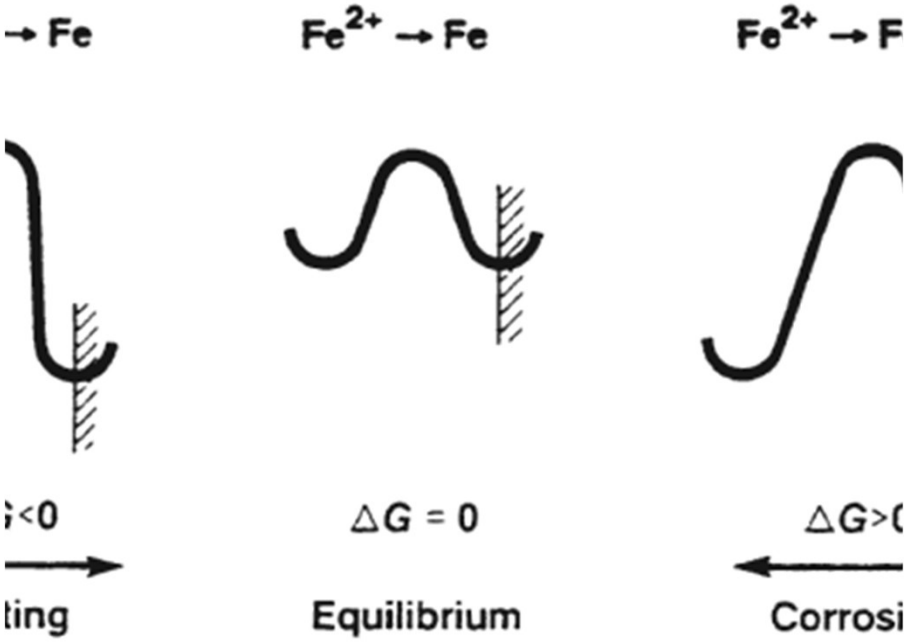
Microbial Colonisation…..

- Microbial cells arriving near the implanted surface
  - Direct contamination
  - Transcutaneous spread
  - Seeding from infected fluids flowing over the device

- Are drawn near to the surfaces by forces
  - Hydrophobic interaction
  - Electrostatic attraction

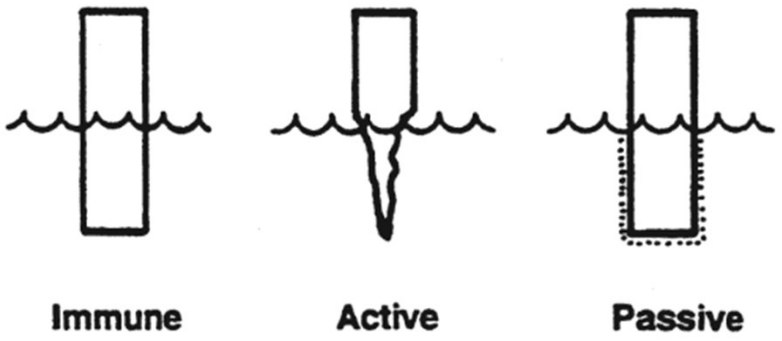
Why corrosion happen?

- A metal in contact with a solution moves towards the lowest free-energy state.



Behavior of Metal in an environment

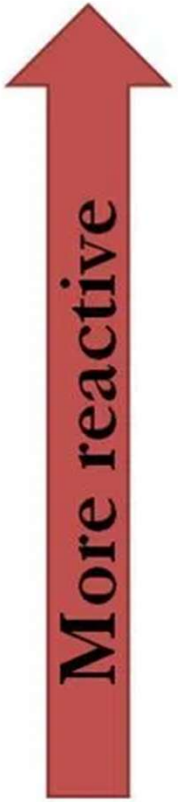
- Immune
  - Metal is immune to environment - No corrosion - Noble Metal (Gold, Platinum)
- Active
  - Metal corrodes in the environment - dissolves.
- Passive
  - Metal corrodes but a state of passive behavior is observed - protective corrosion product film is formed. (Iron, Chromium,Titanium,Nickel)



- Each metal has inherent tendency to corrode
  - The EMF series gives a ranking of the metal with respect to their inherent reactivity.



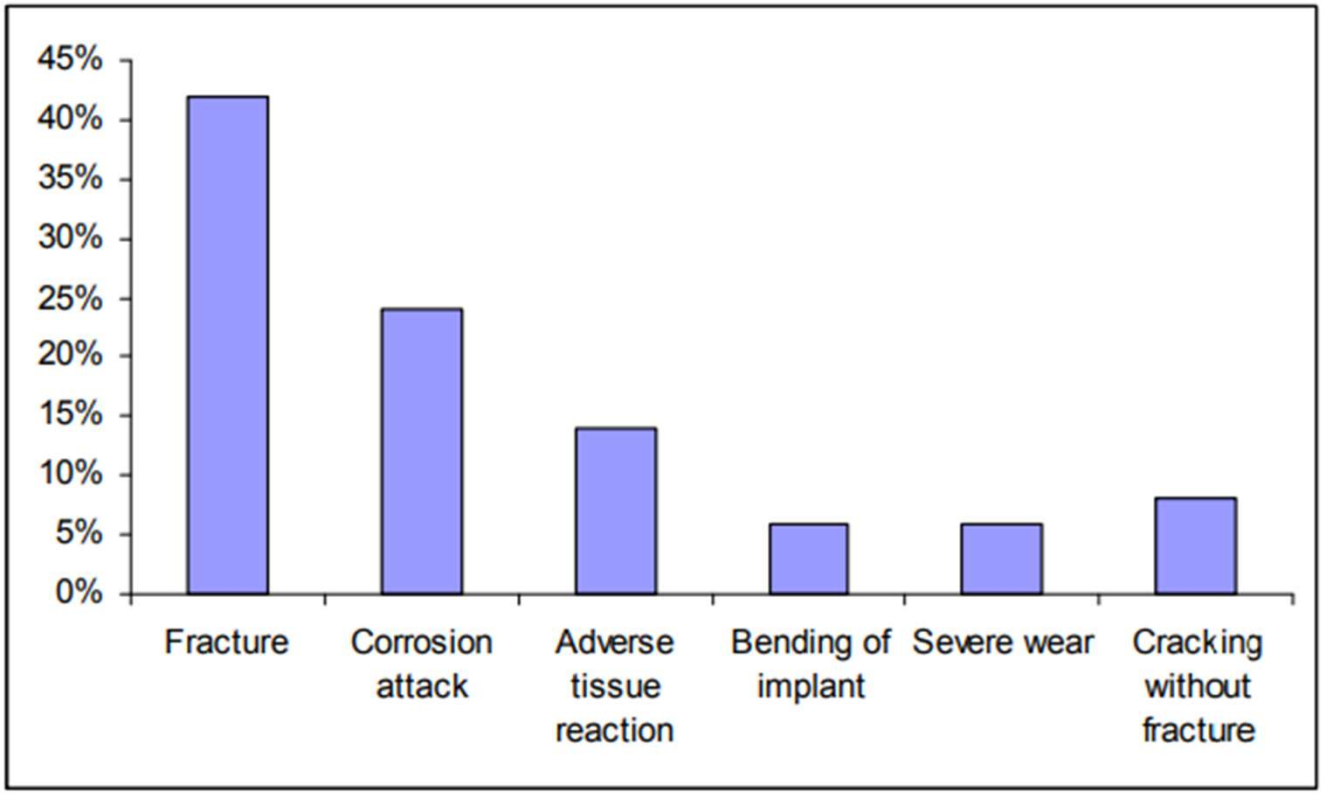
Element	Reduction potential / V
Lithium	-3.04
Potassium	-2.936
Calcium	-2.868
Sodium	-2.714
Magnesium	-2.36
Aluminum	-1.677
Lead	-1.26
Zinc	-0.762
Iron	-0.44
Cobalt	-0.282
Nickel	-0.236
Tin	-0.141
Copper	0.339
Silver	0.799
Mercury	0.852
Platinum	1.18
Gold	1.69



Solution characteristics that affect corrosion

- Directly proportional to *conductivity* of solution.
- pH of the solution
  - Depends on the metal & the solution.
- Oxidising power of solution.
  - Solution of high oxidising power corrode all the metals except metals have most +ve values in emf series.
- Corrosion rate is directly proportional to the concentration of oxygen in the solution.

Causes of failure of Stainless steel Implants



Why metals corrode in human body ?



The aqueous medium in the human body is highly corrosive due to the presence various ions such as chloride, phosphate, and bicarbonate ions, cations like Na^+ , K^+ , Ca^{2+}, Mg^{2+} etc.



The concentrations of chloride ions in serum and interstitial fluid : 113 and 117 mEq/L, makes it a VERY corrosive environment for metallic materials.



The change in the pH(5.3-5.6) leads to, producing highly reactive oxygen species. These oxidizing species (e.g. H_2O_2) will interact with the implant surface.

Why metals corrode in human body ?

- Biological molecules upset the equilibrium of the corrosion reaction - consuming the products of anodic and cathodic reaction.

- Protein binds to metal ions, transport them away, unsets the protective surface layer.

- Hydrogen formed in the corrosion reaction - corrosion inhibitor. Certain bacterias absorb hydrogen - Enhance corrosion

Standards for Testing Corrosion Resistance of Biomaterials

ASTM Standards	Specifications
ASTM G 61-86, and ASTM G 5-94	Corrosion performance of metallic biomaterials
ASTM F746-87	Pitting or crevice corrosion of metallic surgical implant materials

Tolerable corrosion rate for metallic implants : *2.5 * 10^(-4) mm/yr*

Effects of Corrosion in Human Body Due to Various Biomaterials

Biomaterial Metals	Effect of Corrosion
Nickel	Affects skin - such as dermatitis
Cobalt	Anemia B inhibiting iron from being absorbed into the blood stream
Chromium	Ulcers and Central nervous system disturbances
Aluminum	Epileptic effects and Alzheimer's disease
Vanadium	Toxic in the elementary state

Methods for corrosion alleviation

- Surface Modification.
  - Diamond-like carbon (DLC) coatings - Excellent Corrosion resistance, Excellent Hemocompatibility.

- Corrosion performance of medical devices made of stainless steels can also be improved by electrochemical polishing - removes material - make surfaces smooth.

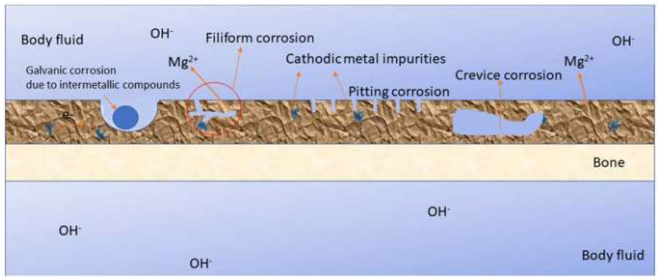
- Use of Ceramics - Alumina and zirconia are considered to be as alternatives for metallic materials for load bearing applications. etc.

Types of bioimplants corrosion

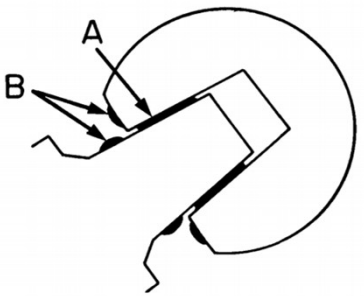


Pitting corrosion

- Pitting is a severe form of localised corrosion attack
- Results in extensive damage and release of metal ions
- Pitting refers to the formation of small cavities/holes at the surface of a material
- The formation of pits is due to the interaction of certain aggressive ions with the film at locations where it is defective or weak in nature
- They can allow the formation of stress corrosion cracking(SCC) or fatigue cracks



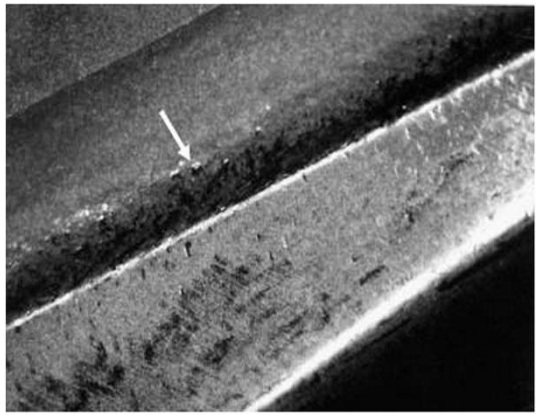
Types of bioimplants corrosion….



A = thin interfacial layer of mixed oxides and chlorides within the crevice formed by the junction of the head and the neck taper.
B = corrosion products

Crevice corrosion

- Related to structural details occurs when a metal surface is partially shielded from the environment
- It is usually encountered beneath the screw head that holds the plate
- The basic requirement for the occurrence of this process
- Presence of a crevice, a narrow, deep crack
- Type 316L stainless steel is highly susceptible
- Can be eliminated by appropriate design of device and proper choice of the material



Localized corrosion on a cemented Ti-6Al-4V stem region due to crevice attack.

Types of bioimplants corrosion….

Galvanic corrosion

- Occurs when two different metals are in physical contact in an ionic conducting fluid medium (serum or interstitial fluid)
- Galvanic corrosion depends
  - The relative areas of electronic and ionic
  - Contact and actual metal pair involved

- In surgical implants, can occur if a bone plate and bone screw are made of dissimilar metals or alloys



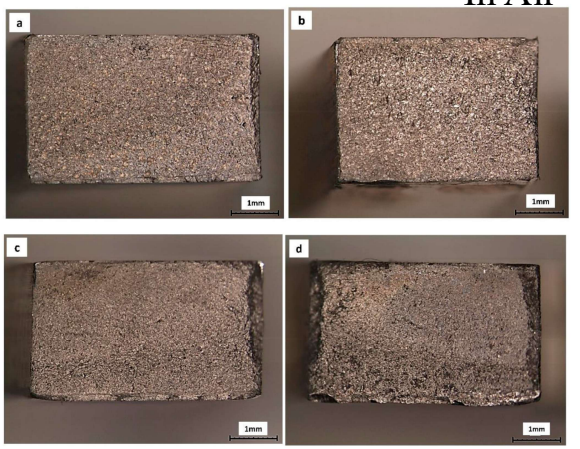
Between Head-Neck

Types of bioimplants corrosion…..

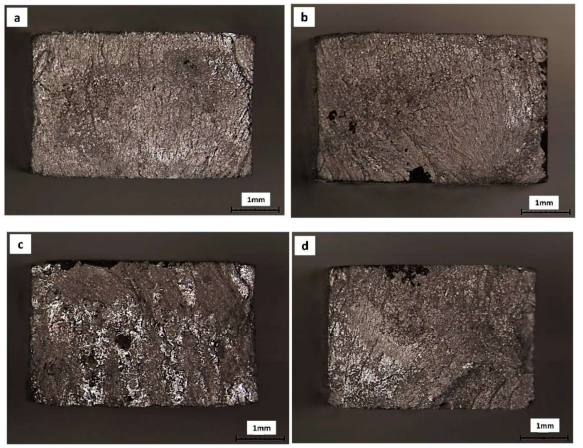
Corrosion fatigue

- Is a fracture failure of metal that occurs because of the combined interaction of electrochemical reactions and cyclic loading
- Corrosion fatigue resistance is an important factor of consideration for load-bearing surgical implant metals/ cyclic motion applications
- Cracks could be initiate because of corrosion phenomena

In Air



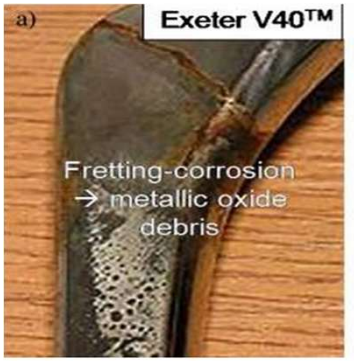
In PBS

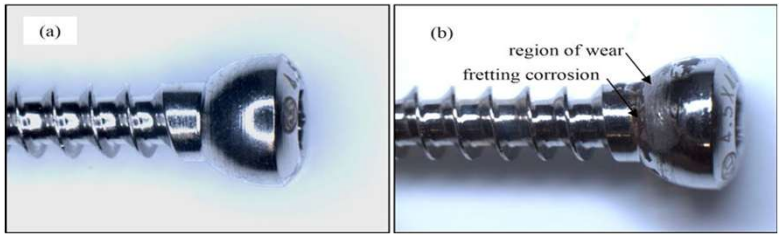


Types of bioimplants corrosion…..

Fretting corrosion

- Opposing surfaces (bone plates, the screw heads) of the prosthetic devices rub each other continuously in an oscillating fashione
- May give rise to corrosion products in adjacent tissues or it may be a major factor in crack initiation and fracture failure of an implant
- Fretting corrosion at countersinks in plates and at hip nails may initiate corrosion fatigue (which occurs through screw hole)





Occurs at the interface for two contacting surface due to small oscillatory movements arising between them in the presence of a corrosive medium.

Effects of corrosion on tissues (Toxicity)

- The corrosion of metallic implants can affect the surrounding tissues in three ways
- Electrical current may affect the behaviour of cells
- The corrosion process may change the chemical environment
- Metal ions may affect cellular metabolism

Effects of corrosion on tissues…

- Issues that arise from the release of corrosion products into the body is
- **Remote effects**; local tenderness at the site of the corroded area to acute pain, reddening and swelling over the whole general area around the device

- **Systemic**; elevated concentrations of specific alloying elements of implants (Organ specific accumulations)
- This could upset the overall balance established by physiological tolerance to toxicity

Effects of corrosion on tissues…

Metallosis *;* local destruction of soft and hard tissues due to metal implants

- Is associated with significant osteolysis
- Is sometimes reflected by an infiltration of periprosthetic soft tissues and bone by metallic debris resulting from wear of joint arthroplasties
- By themselves, metal ions lack the structural complexity required to challenge the immune system
- Their interaction with body tissues induce the immune responses (should be considered harmful)

Effects of corrosion on tissues…

The following factors must be considered when dealing with toxicity of metallic biomaterials

- Corrosion resistance of the material
- Ions released by corrosion and wear
- Activity of the released ions
- Toxicity of the ion itself

Thanks For Your Attention